higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## MULTIMEDIA SERVICES
## NQF LEVEL 4

## 8 MARCH 2019

**This marking guideline consists of 8 pages.**

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## SECTION A

## QUESTION 1

| | | |
|---|---|---|
| 1.1 | 1.1.1 | B |
| | 1.1.2 | A |
| | 1.1.3 | A |
| | 1.1.4 | A |
| | 1.1.5 | C |
| | 1.1.6 | C |
| | 1.1.7 | D |
| | 1.1.8 | D |
| | 1.1.9 | D |
| | 1.1.10 | B |
| | 1.1.11 | A |
| | 1.1.12 | B |
| | 1.1.13 | B |
| | 1.1.14 | D |
| | 1.1.15 | A |

(15 × 1) (15)

| | | |
|---|---|---|
| 1.2 | 1.2.1 | Hierarchical database |
| | 1.2.2 | Database |
| | 1.2.3 | Encryption |
| | 1.2.4 | Table |
| | 1.2.5 | Database administrator (DBA) |
| | 1.2.6 | Primary key |
| | 1.2.7 | Index |
| | 1.2.8 | Authentication |
| | 1.2.9 | Object-oriented |
| | 1.2.10 | Database management system |
| | 1.2.11 | Differential backups |
| | 1.2.12 | Interface |
| | 1.2.13 | Foreign key |
| | 1.2.14 | Distributed database |
| | 1.2.15 | Full backup |

(15 × 1) (15)

| | | |
|---|---|---|
| 1.3 | 1.3.1 | False |
| | 1.3.2 | True |
| | 1.3.3 | True |
| | 1.3.4 | True |
| | 1.3.5 | False |
| | 1.3.6 | True |
| | 1.3.7 | True |
| | 1.3.8 | True |
| | 1.3.9 | True |
| | 1.3.10 | False |
| | 1.3.11 | True |
| | 1.3.12 | False |
| | 1.3.13 | True |
| | 1.3.14 | False |
| | 1.3.15 | False |

(15 × 1) (15)

1.4 1.4.1 ISDN – Integrated Services Digital Network
1.4.2 ADSL – Asymmetric digital subscriber line
1.4.3 IIS – Internet Information Services
1.4.4 DNS – Domain Name System
1.4.5 HTTPS – Hypertext Transfer Protocol Secure

(5 × 1) (5)
**[50]**

**TOTAL SECTION A: 50**

## SECTION B

## QUESTION 2: NETWORKS AND SERVERS

2.1 Cables that are too long would result in signal loss.✓ This means that data transmission and reception would be affected,✓ because the signal degrades over length.✓

(3)

2.2
- When one workstation on the network suffers a malfunction, it can bring down the entire network.
- Adjustments and reconfigurations need to be performed on a particular part of the network; the entire network has to be temporarily brought down as well.

(2)

2.3 IPv6, or Internet Protocol version 6, was developed to replace IPv4. At present, IPv4 is being used to control internet traffic and allow user's access to the world wide web. ✓ Each continent was allocated a certain amount of addresses to use and in the near future we will be running out of IP addresses to allocate to new users. IPv4 can only make available approximately 4.3 billion IP addresses. IPv6 was designed to overcome this limitation.✓ IPv6 can make available approximately 3,4 × 10^38 or 340 undecillion IPv6 addresses. ✓

(3)

2.4 This label normally means that the ports on the switch are capable of operating at 10 Mbps or 100 Mbps.✓ Some older switches have 10 Mbps ports for connecting systems and 100 Mbps ports for up-linking. ✓ Because there are no guidelines for labelling devices✓, some of those older switches are just referred to as 10/100 switches.✓

(4)

2.5 Bridges are networking devices that connect different networks with the same protocol.✓ Sometimes it is necessary to divide networks into subnets to reduce the amount of traffic on each larger subnet or for security reasons.✓ Once divided, the bridge connects the two subnets and manages the traffic flow between them.✓ A bridge functions by blocking or forwarding data, based on the destination MAC address written into each frame of data. ✓If the bridge believes the destination address is on a network other than that from which the data was received, it can forward the data to the other networks to which it is connected.✓ If the address is not on the other side of the bridge, the data is blocked from passing. ✓ Bridges 'learn' the MAC addresses of devices on connected networks by 'listening' to network traffic and recording the network from which the traffic originates.✓ The bridge will be placed at a

suitable location so as to join the 2 networks together. ✓

(8)

**[20]**

## QUESTION 3: DATABASE MANAGEMENT

3.1 CREATE DATABASE✓✓*Hospital*✓ (3)

3.2
```
CREATE TABLE Patients (✓
  Patients ID varchar(50),
  Surname varchar(50),
  Address varchar(50),
  Fees  int,
  CellNo varchar(20);
);✓
```
✓ **for listing all correct columns with var lengths** (3)

3.3.1
```
SELECT Surname, Fees✓
FROM Hospital✓
WHERE Fees > 2300✓✓
```
(4)

3.3.2 The primary key in this tabel is: **PatientID** .✓
The alternate Primary Key is : **CellNo.** ✓ (2)

3.3.3
```
Select ✓ * ✓
From✓ Patients      ✓
```
(2 × 2) (4)

3.3.4
```
DELETE✓ FROM Patients✓
WHERE PatientsID✓=2008128792  ✓
```
(2 × 2) (4)

**[20]**

## QUESTION 4: WEB PLANNING

4.1
- Blinking, spinning or flashing images
- Black background with white, light or pale text
- Black backgrounds with dark text
- Busy tiled background images with any colour text
- Everything cantered
- Too many images or huge images
- List of links
- Too many headlines or blinking text

(Any 6 × 1) (6)

4.2 A marquee tag is used when some text or content has to move on the web page ✓either horizontally or vertically. ✓

Syntax of marquee tag:
<marquee>move text</marquee>✓will move the text from left to right ✓
<marquee direction = "down"> ✓move text</marquee> will move the text up and down✓
✓✓ for explanation
✓ for each syntax used✓ for an explanation (6)

4.3 ADVANTAGES OF A RING TOPOLOGY

1. Reduced chances of data collision✓ as each node releases a data packet after receiving the token.✓
2. Token-passing makes ring topology perform better✓ than bus topology under heavy traffic.✓
3. No need of server to control connectivity among the nodes.✓ (6)
4. Equal access to the resources.✓ (Any other correct answers) (3 × 2)

4.4 This first web browser, which was named 'WorldWideWeb,' had a graphical user interface.✓ ✓ (2)
**[20]**

## QUESTION 5

5.1
```
<!DOCTYPE html>✓
<html>
<head>
   <title>My First Website</title>✓
        <link rel="stylesheet" href="nicestyle.css" type="text/css" />✓
</head>
<body>
<div id="page">
  <div id="logo">
  <h1>My First Website</h1>✓
       </div>
       <div id="nav">
       <ul>
           <li><a href="#/home.html">Home</a></li>✓✓
```

```
            <li><a href="#/about.html">About</a></li>
            <li><a href="#/contact.html">Contact</a></li>
          </ul>
                </div>
                <div id="content">
                 <h2>Home</h2>✓
           <p>
            This is my first webpage. I was able to code it all in HTML <br>
        and CSS in order to make itlooklike this. Watch out!!!<br>
        Here I come with my Design!<br>
          </p>
            <p>
                 I can use my <b>coding skills to create websites<b> for business, <br>
             also for my friends and family. <br>
           </p>
                </div>
                <div id="footer">✓
            <p>
              Webpage made by <a href="/" target="_blank">[NCV Student]</a>✓✓
            </p>
                  </div>
          </div>
</body>
</html>
```

(10)

5.2

```
<html>
    <body>
        <script type = "text/javascript">✓
            document.write("Welcome to my Home Page");✓
        </script>
    </body>✓
<html>
```

(3)

5.3

```
// I like website Programming
/* I like website Programming*/✓
<!—I like website Programming -->✓
```

(2)

5.4

```
<html>
   <head>✓
          <title> CSS Styling </title>✓✓
          Link rel = "stylesheet" href = "nicestyles.css" type = "text/css"/>✓✓
    </head>
```

(5)
**[20]**

## QUESTION 6

6.1 1. TO MAKE MONEY✓
Malware is designed to capture information about the computer user and send it to the

person or company responsible for making the malware. The information they collect is then used to target advertisements to your computer that you may be interested in viewing. If enough computers get infected, they can earn money from all the ads displayed. ✓

2. TO STEAL ACCOUNT INFORMATION✓
   A person could gain access to a victim's account and steal their virtual goods and currency. They could even sell their ill-gotten virtual goods to other players for real money. ✓

3. TO CAUSE PROBLEMS AND TROUBLE FOR OTHERS✓
   On the other side, there are those people that create viruses and malware because they can. They enjoy causing trouble, seeing computer users get upset and have to fix their computer. Some malware can crash an entire network system and cause system outages for large companies, like banks or production companies. ✓

(Any other suitable explanation the student may give)

(6)

6.2

| FIREWALL | ANTIVIRUS |
|---|---|
| It is a hardware/software based network security system that prevents unauthorised access between computer programs. ✓ | Is a software utility program designed to safeguard your computer from internal threats such as viruses, malware, spyware, etc. ✓ |
| It inspects the data flow from the internet to the system in order to decide which one should be allowed and which one should be restricted✓ | It detects scans, prevents, and removes harmful programs that might compromise your computer. ✓ |
| Firewall can also be referred to as a "packet filter". It acts as a barrier between your computer and the network by monitoring incoming and outgoing traffic and analyzing the data packets that travel along the network. Basically it filters the packets before sending it through or else if found suspicious, it discards the packets. ✓ | Is a set of programs designed that will find vulnerabilities in your system and take the necessary action. ✓ |
| A firewall usually works at the network protocol level to safeguard both public and private networks from unwanted intrusion. It can only prevent unauthorized access between computer networks based on a pre-defined set of network protocols. ✓ | Antivirus works at the file level meaning the software program will only scan for harmful programs that are installed on your system such as worms, Trojans, adware, spyware, malware, rootkits, keyloggers, etc. ✓ |

(8)

6.3 An IDS is a sensor✓ like a smoke detector that raises an alarm if specific things occur. Intrusion detection is the process of identifying and responding to malicious activities targeted at computing and network resources.✓ It involves technology, people and tools. An intrusion detection system basically monitors and collects data from a target system that should be protected, processes and correlates the gathered information and initiates responses when an intrusion is detected.✓

(3)

6.4 The IPS (intrusion prevention system) needs to see the traffic it is protecting and shouldn't be in the way of traffic it isn't protecting.✓

Your IPS will generally be placed at an edge of the network, usually immediately inside an internet firewall, or just in front of a server.✓ Position the intrusion prevention system where it will see the minimum of traffic that it needs to, in order to keep the network performance issues under control.✓

(3)
**[20]**

**TOTAL SECTION B: 100**
**GRAND TOTAL: 150**

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